A Replication of Geographic Dispersion of Economic Shocks: Evidence from the Fracking Revolution

Feyrer, Mansur, and Sacerdote (AER 2017)

Quan Nguyen

ECON 6901: Research Methods 1

Dr. Paul Gaggl & Dr. Craig Depken

December 10, 2025

I. Introduction

During the mid 2000s and early 2010s, the advancements of 3-D seismic imaging, horizontal drilling, and hydraulic fracturing created an unexpected oil and natural gas boom in the US. The interaction of these technologies with preexisting shale rock formations resulted in massive sudden production of gas and oil across the country and propelled the US into global energy stardom. This paper takes the fracking revolution as a natural experiment to study how energy shocks affect income and how those effects evolve over space and time. Understanding the distribution and persistence of the income effect not only gives operational guidance to policymakers, but it is crucial for assessing whether resource shocks generate broad or concentrated growth.

By using a detailed panel of US counties from 2005 to 2012 and combining new well production data with BLS and IRS income data, I estimate the causal effects of fracking on income. More specifically, I ask: how much of the value of new fracking production stays within the wellhead county and its surrounding region, how do these effects change over time, and which industries see benefits? The income effect from fracking can take several forms: workers in fracking related industries, workers in non-related sectors, and land/mineral owners who receive royalties on the value of new production.

Two main results, which are consistent with the original study, emerge from the analysis. First, fracking counties see a dramatic positive income effect. Within the county, each one-million-dollar worth of new production increases wage income by $80,000 and royalty and business income by $132,000, implying that roughly 21% of the production value remains in the wellhead county. Within 100 miles of the county, each one-million-dollar of new production increases wage income by $257,000 and royalty and business income by $286,000, implying that 54% of production value remains within 100 miles. These income gains are also persistent: roughly two-thirds of the wage income increase persists for two years, while royalties are short lived, implying that benefits are not entirely fleeting. We also see that the income wage composition shifts from extraction related industries in the impact year, to non-related sectors within the 2 years of the impact, implying a broader labor market effect.

Econometrically, I estimate two-way fixed effects and IV models that use geological shale presence to isolate exogenous variation in production. This method captures changes driven by the timing of technology and geology rather than local economic conditions that may be endogenous. I extend this analysis by examining both the spatial propagation of these shocks and temporal persistence of the effects, providing a better economic understanding of how a resource shock impacts local economies over space and time.

I. Literature Review

Natural resources and the sudden influx of them have prompted economists to study how local and regional economies respond to such shocks. An ongoing debate in local labor economics is whether these fossil fuel booms generate broad-based gains or a resource-curse in which wages in one sector rise while others are crowded out.

On one side, boom-bust evidence shows asymmetric losses after its initial positive effects on employment and income, especially in the manufacturing industry (Black, McKinnish and Sanders, 2005; Jacobsen and Parker, 2016). In contrast, other studies find overall benefits to regional economies: resource shocks can expand the manufacturing industry, raise non-mining employment, and increase income over time (Allcott and Keniston, 2014; Weber, 2013; Fetzer, 2014; Michaels, 2011). The paper sheds light on this central question by causally estimating the effects of fracking on wages and salaries and assessing the differences across sectors and regions.

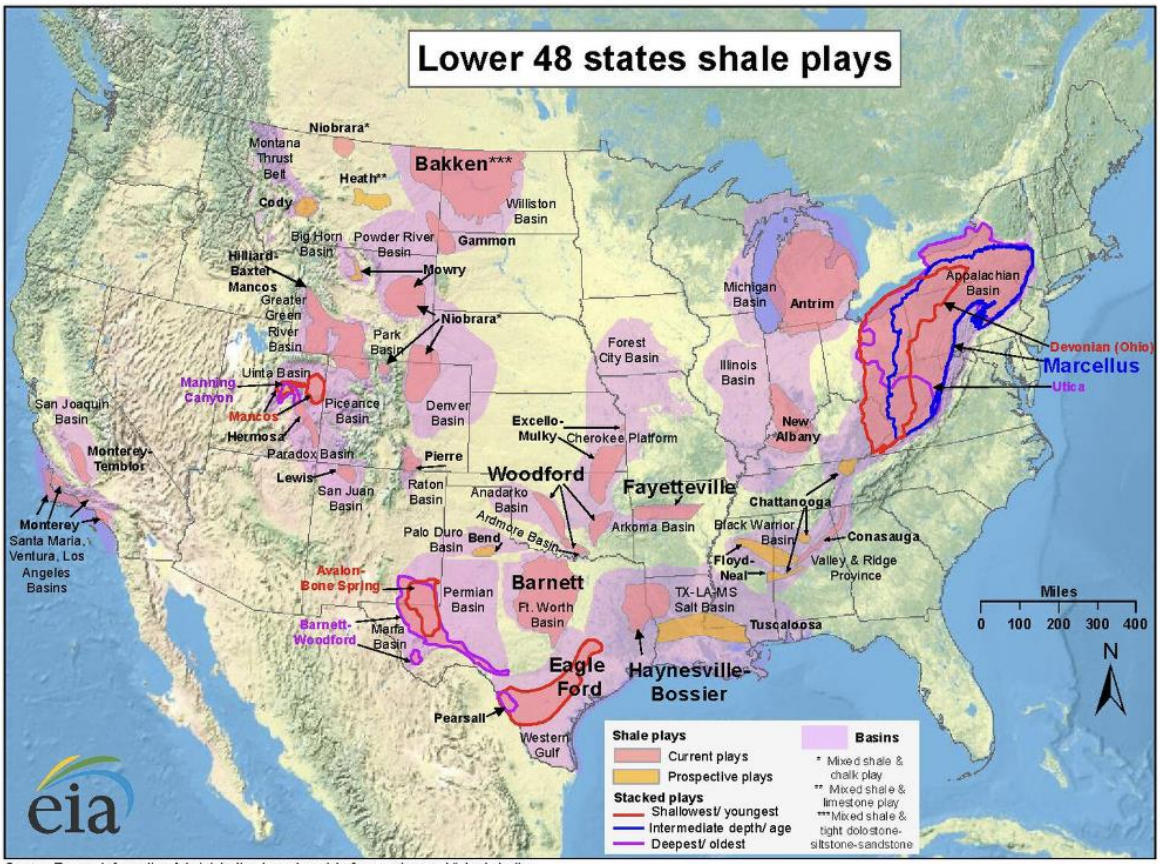
The fracking revolution offers significant variation in the study of resource shocks because the interaction of fixed geology with new extraction technology generated large, uneven new production across space and time. Early studies on fracking focused on particular states or sectors (Weber, 2012) or used indirect proxies for measuring production such as shale presence or the number of wells (Fetzer, 2014; Maniloff and Mastromonaco, 2014). This paper differs by using detailed measures of the value of production from new wells for all 48 contiguous states, which better aligns the shock with when and where the activity takes place. This larger, detailed measurement method enables us to find clearer evidence on which industries capture wage and employment gains, and how these effects propagate over space and time.

Local shocks, in general, rarely stop at jurisdictional borders. In the case of fracking, many counties often share a shale play, and the effects are often distributed as supplier networks relocate activity and workers migrate or commute across borders, which is why it is important to estimate responses over specific regions, rather than just at the county level, to capture these spillover effects (Blanchard and Katz, 1992). Ultimately, by treating the fracking revolution as a large and arguable exogenous shock, we can causally estimate how income and employment respond to the shock, which industries benefit, how far the effects spread, and how long they persist (Shoag, 2010).

II. Local Geology and Technology of Fracking

Shale oil and natural gas are fossil fuels trapped in low-permeability shale rock formations, commonly referred to as shale plays. Although these shale plays were known to hold these resources for decades, they remained largely unextractable until the early

2000s, when the combination of high-pressure hydraulic fracturing, horizontal drilling, and advanced seismic imaging made extraction on these shale plays technologically feasible. These developments led to a massive expansion of U.S. oil and gas output, contributing to significant declines in domestic energy prices. More importantly, production is not geographically uniform as shale reserves are concentrated in specific rock formations that vary greatly in depth, thickness, brittleness, and natural fracture patterns. The map below illustrates the location of major shale plays across the U.S., which helps clarify the location of the resources and where fracking activity became possible.



Source: Energy Information Administration based on data from various published studies.
Updated: May 9, 2011

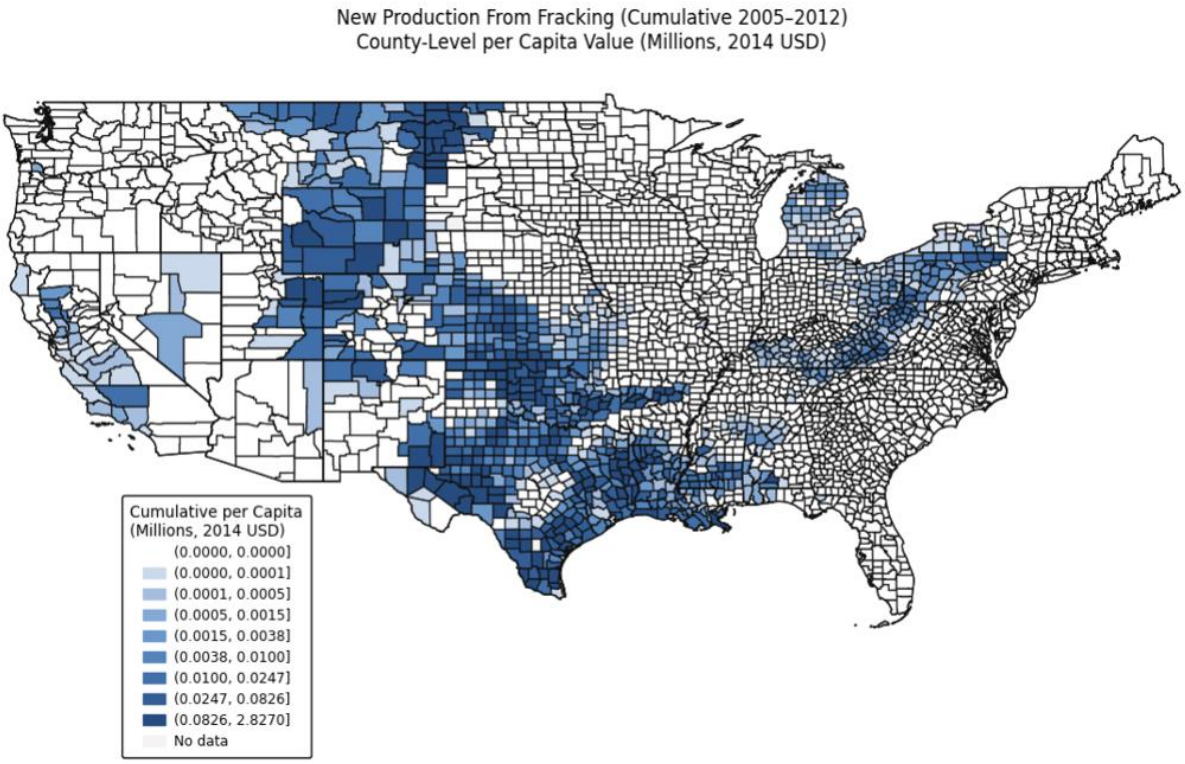
For our identification, it is critical to understand that the timing of adoption differed sharply across plays. Some shale formations were geological sweet spots, which made them immediately viable for fracking, whereas others required years of experimentation and learning. In addition to rock characteristics, timing varied across plays for other reasons, such as commodity prices, pipeline infrastructure, permitting rules, service-

sector capacity, and access to inputs like water and proppant sand. This heterogeneity caused certain plays, such as Barnett and Fayetteville, to expand early, while others, such as Utica and Wolfcamp, took off only much later. As a result, the fracking boom did not arrive as a single national shift, but rather as a staggered series of play-specific expansions. This staggered timing is central to the IV strategy: we construct instruments using data on the presence of shale in each county's underlying geological formation from the EIA and assign each county to a specific play, grouping small plays with fewer than ten counties into a single "other" category. Because counties are geologically predetermined in their exposure to particular plays, the year in which a play becomes economically frackable provides plausibly exogenous variation in the timing and magnitude of new production across counties.

III. Data

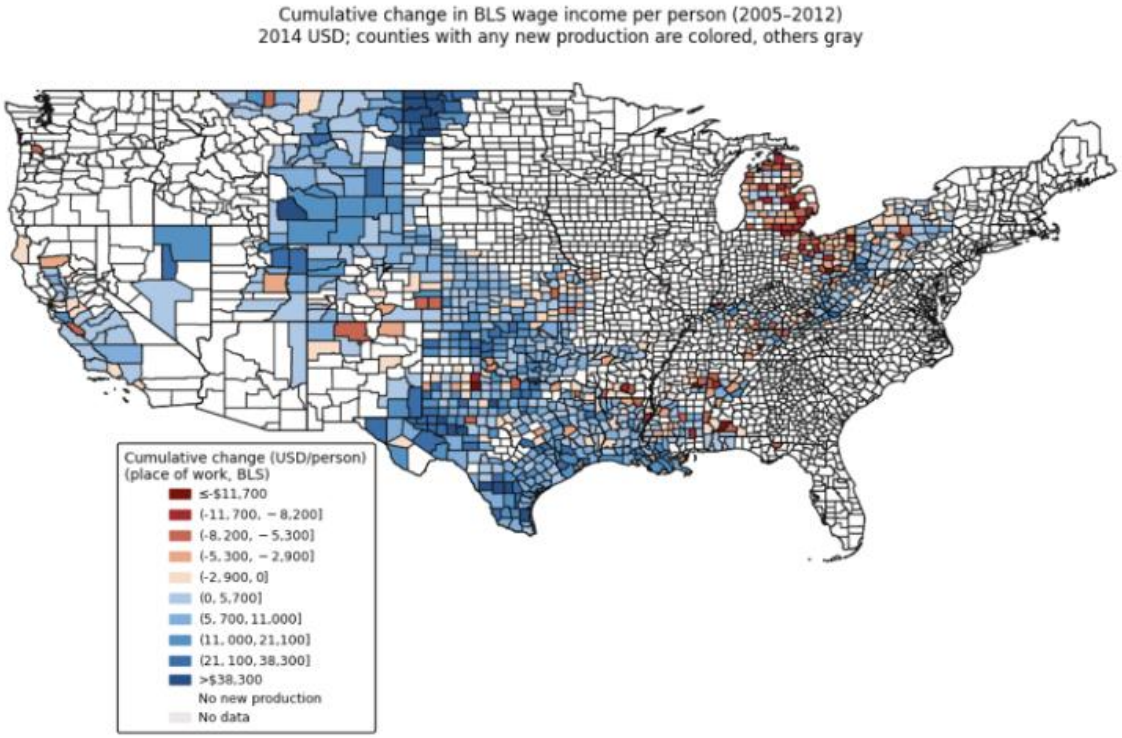
The authors obtain well-level production data from DrillingInfo, which reports monthly oil and natural gas output for each well in each county of the United States. To construct our production measure, we value the oil and gas output using benchmark energy prices from the EIA and aggregate it to the county-year level for analysis. The study focuses exclusively on new production, which is defined as the value of output from wells that begin producing in a year t. This choice isolates the timing of the economic shock (when a well becomes active) rather than capturing ongoing output from sites that have already been fracked. Thus, the value of new production serves as our primary measure of the local economic shock. Most of the new oil production during this period is concentrated in California, Colorado, Kansas, Louisiana, New Mexico, North Dakota, Ohio, Oklahoma, and Texas, while new gas production is concentrated in Arkansas, Colorado, Louisiana, New Mexico, Pennsylvania, Utah, Texas, West Virginia, and Wyoming. The paper excludes the smallest two percent of counties in terms of total employment to avoid extremely large outliers driven by counties such as Loving, Texas, that produced hundreds of millions of dollars of new oil in certain years, but employed fewer than fifty workers

The figure below is a replication of figure 1 from the original paper, mapping each county's cumulative value of new production from 2005 to 2012, measured in 2014 millions of dollars. At a glance, drilling is not entirely random—only 25% of counties in the U.S. see any drilling at all from 2005 to 2012. Among the counties that do see any drilling, the amount of fracking varies greatly across space, with a lot of clustering in the dark colored shale plays like Eagle Ford, Bakken, Barnett, etc. Such spatial heterogeneity highlights the importance of including county fixed effects, which removes time-invariant differences (persistent characteristics) across counties.

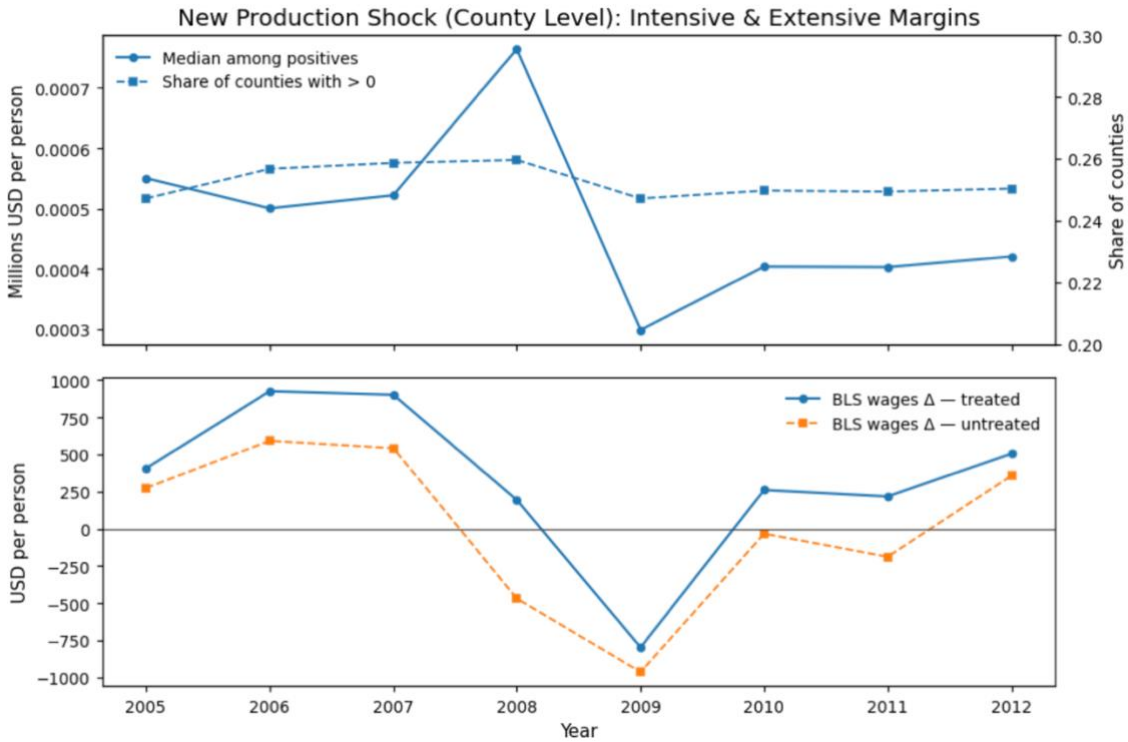


Furthermore, the authors obtain county-level annual average total wages and employment data, along with industry-level decompositions, from the Bureau of Labor Statistics' Quarterly Census of Employment and Wages. This data originates from unemployment insurance records, where employers report quarterly wages and employment information. More importantly, BLS wages are recorded in the county where the employer is located, regardless of where the workers live or where work is physically performed. As a result, BLS wages more accurately capture where economic activity occurs than IRS wage data, which reflects workers' counties of residence. Because the BLS series is disaggregated by industry, we can study whether the income effects from fracking are concentrated in certain extraction-related industries or whether they spill over into non-related sectors.

In addition to the original paper's map of the cumulative value of new production, I constructed a similar map, shown below, that plots each treated county's cumulative change in BLS wages per person from 2005 to 2012, in 2014 millions of dollars. Among the counties that see fracking, cumulative wage income generally rose, but not uniformly across space. This nonuniform pattern is consistent with the earlier map and reinforces the use of county fixed effects.



A second descriptive fact that I add comes from decomposing fracking shocks into their extensive and intensive margins. The top panel of the figure below shows the share of counties that receive any new production in each year (extensive), along with the median value of new production among counties that do see a new well (intensive). The bottom panel plots the median annual change in BLS wage income per person for treated and untreated counties. From this comparison, treated and untreated counties follow a very similar time pattern in wage movement, with the difference being that treated counties generally exhibit higher wages than untreated counties. Moreover, the pattern of wage movement varies across years, and macroeconomic downturns such as the 2008 recession affects treated and untreated counties alike. For example, from 2007 to 2008, the median new production shock among drilling counties rises, yet the median wage essentially falls for all counties. Without controlling for time effects, we could wrongly attribute recession-driven wage declines to an increase in drilling. This motivates the use of year fixed effects, which remove nationwide shocks common to all counties in each year. In combination with county fixed effects, our identification comes from county-year deviations that are above or below both the county's own average and the year's national level.



The authors also use data from the IRS Statistics of Income, which has information on annual AGI, wages and salaries, and capital gains for each county-year. We are not particularly interested in IRS wages, since they reflect where the workers live and not where the economic activity occurred. However, we define IRS other income as the difference between AGI and wages, dividends, interest, and capital gains. The IRS other income categories capture royalties to mineral right holders and landowners, who are paid significantly by corporations interested in using their land for fracking. More specifically, these royalty payments make up between one-eighth and one-fourth of the total value of production (Urbina 2011, Fitzgerald 2014). The IRS other income, therefore, captures the household-level channels (royalties and business income) through which fracking activity benefits residents beyond wages paid in the drilling county. This category also includes non-wage business income like rental income.

Following the original study, I exclude the smallest two percent of counties from the county-level analysis to avoid extreme per-capita values driven by very low-population counties that generate disproportionately large production per worker. For example, in 2007, Loving County, Texas, employed only 44 workers and produced roughly $761 million

in new oil per capita. However, these counties are retained for summary statistics and for analyses that aggregate beyond the county level (e.g., distance-based results).

IV. Empirical Framework

A. Econometric Model

Following Feyrer, Mansur, and Sacerdote (2017), the empirical goal is to estimate how much economic value is generated by new oil and natural gas production and how much of that value remains within the local labor market. The analysis begins by relating annual changes in county-level outcomes, ΔY_{it}, such as BLS wages or IRS other income, to the value of new production, $NewValue_{it}$, all scaled by lagged employment. Scaling by employment makes drilling intensity comparable across counties of different population sizes and prevents the estimated effects from being driven by population levels. I estimate a baseline specification that includes county fixed effects a_i , which absorbs time-invariant characteristics such as baseline economic composition and long-run industry mix, and year fixed effects ω_t, which capture nationwide business cycle conditions and macroeconomic shocks common to all counties. Consistent with the paper, I also include a one-year lag of new production, $NewValue_{i,t-1}$, to control for potential dynamic effects. Thus, for a county *i* in year *t*, I estimate:

$$(1)\ \Delta Y_{it} = \beta_1 \times NewValue_{it} + \beta_2 \times NewValue_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

Here, β_1 captures the contemporaneous income response (BLS wages or IRS other income) to new production, conditional on lagged new production and the set of fixed effects. Standard errors are two-way clustered by county and year to account for serial correlation within counties and spatial correlation within a year. This approach adjusts the standard errors appropriately when shocks propagate across space and persist over time.

B. Spatial Propagation

Although counties are the natural unit of observation, the economic activity generated by fracking is unlikely to be confined to the wellhead county where extraction physically occurs. Drilling often takes place in sparsely populated areas, so workers may reside and consume elsewhere in neighboring counties, and employment associated with extraction may be recorded by the BLS where the supervising firm is located rather than where drilling takes place. To capture these broader regional effects, I follow the original study and aggregate both the value of new production and income within increasing radii around each county's geographic centroid. Specifically, for each county i, I sum the variables and employment for all counties lying within R miles of i and construct regional

production per worker by dividing the aggregated variables by aggregated employment and re-estimate the baseline model at radii ranging from 20 to 200 miles. Comparing the estimated coefficients across each radii traces how economic benefits diffuse across space, revealing that the regional impact rises as the geographic area widens, with most of the income effect occurring within 100 miles of the wellhead county. This spatial aggregation approach shows that the fracking boom not only benefits the wellhead counties but also generates spillovers to the surrounding counties.

C. Instrumental Variables

While the geological location of shale plays and the availability of resources are predetermined, the timing of drilling at a particular site may be endogenous if firms choose to drill when local economic conditions are favorable. For example, a firm may choose to drill in counties where labor or land costs are relatively low, where infrastructure already exists, or where certain local downturns lower the opportunity costs of acquiring a lease or permit. These are time-varying, county-specific factors that may be correlated with both new production and income, even after controlling for county and year fixed effects, and therefore could bias the OLS estimates. Therefore, I follow the original paper and construct an instrument for new production that exploits geological exposure to shale plays interacted with the staggered diffusion of fracking technology across plays. The key idea is that different shale formations become economically drillable in different years, so when a particular play becomes commercially viable, counties that sit on top of that play predictably see larger increases in new production. Thus, the IV is constructed by interacting each county's predetermined exposure to a play with the year in which that play becomes economically frackable. Formally, the instrument is built using a two-step procedure with the first step generating predictions for new production for each county-year:

$$(2)\ \ln(NewValue_{it} + 1) = a_i + \lambda_{jt} + \epsilon_{i,t}$$

where a_i captures the county-level geological exposure to a shale play and λ_{jt} denotes the year in which commercial fracking becomes feasible in that play. In the second step, the predicted values from equation (2) are transformed to match the scale of the main independent variable, and the predicted new production per capita for each county-year is constructed as:

$$(3)\ \widehat{NewValue}_{it} = (e^{\widehat{a_i} + \widehat{\lambda_{jt}}} - 1)/pop_{it}$$

This approach isolates the component of new production driven by the staggered diffusion of fracking technology across plays, rather than by contemporaneous local conditions that affect both income and drilling decisions. Thus, the resulting predicted values provide plausibly exogeneous variation in the timing and magnitude of new production across counties and form the basis for the second-stage estimates.

V. Results

A. OLS Results

I begin with a two-way fixed-effects OLS specification at the county level, regressing yearly changes in income per worker (BLS wages and IRS other income) on the value of new production per worker. Using the 2005-2012 panel, the baseline estimates (first table) indicate that one million dollars of new oil and gas production is associated with an increase of $34,000 in BLS wages and $64,000 in IRS other income at the county level. The effect becomes substantially larger as I expand the geographic area of interest: at the county-plus-100-mile level, the associated increases rise to $221,000 for BLS wages and $225,000 for IRS other income, suggesting sizable spillover effects beyond the wellhead county.

However, these OLS results should not be interpreted causally due to bias raised by endogeneity in drilling decisions. Firms may choose when and where to drill based on unobserved local economic conditions (infrastructure, labor markets, input prices) that simultaneously affect income. Consequently, OLS estimates likely distort the true, causal effect of drilling with selection bias and omitted time-varying confounders.

ΔY_{it} = Δ BLS wages	Estimate
β_1	33,957 (10,430)
β_2 (Not S.S.)	4,717 (11,339)

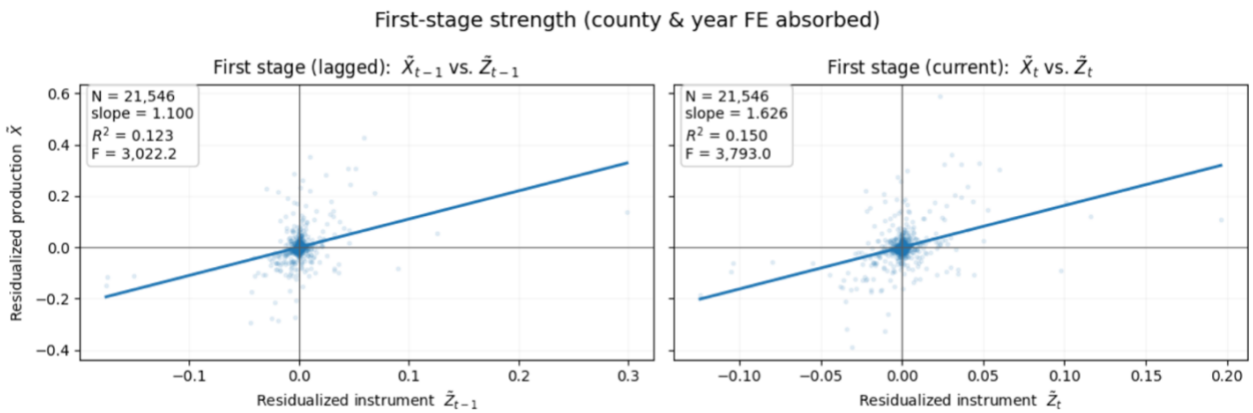
ΔY_{it} = Δ IRS other inc.	Estimate
β_1	64,094 (7,710)
β_2	-43,873 (5,781)

ΔY_{it} = ΔBLS Wages	Estimate
β_1	221,153 (28,256)
β_2 (Not S.S.)	-12,269 (23,747)

$\Delta Y_{it} = \Delta IRS$ other inc.	Estimate
β_1	225,002 (29,375)
β_2	-225,214 (31,784)

B. First-Stage Evidence for Instrumental Variables

To address the endogeneity concerns outlined previously, I employ an instrumental variables strategy based on geological exposure to shale plays interacted with their staggered adoption over time. Before presenting the second-stage IV estimates, it is important to confirm the relevance and strength of the instrument in the first stage. The figure below is a bivariate first-stage scatter plot that plots residualized production per worker against the residualized instrument per worker, where each dot is a county-year observation after demeaning by county and year. A clear positive relationship is visible: counties with greater exposure to a play that becomes commercially viable in a given year tend to experience higher levels of new production.



Formally, first-stage estimates presented below confirm the graphical evidence. The instrument strongly predicts actual new production: the coefficient is large, positive, and statistically significant, and the corresponding first-stage F-statistic is above the conventional weak-instrument thresholds (F>10), even when clustering standard errors by county and year. The lagged instrument similarly strongly predicts lagged new production. Taken together, these findings show that the instrument is relevant, statistically significant, and suitable for use in the second stage.

$$\widetilde{NewValue}_{i,t-1} = \pi_1 \times \tilde{Z}_{it} + \pi_2 \times \tilde{Z}_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

First stage (lagged)	Estimate
π_1	-0.543
π_2	**1.441**

Joint F: 18.4 (p=0.0001)
R^2 = 0.13

$$\widetilde{NewValue}_{it} = \pi_1 \times \tilde{Z}_{it} + \pi_2 \times \tilde{Z}_{i,t-1} + a_i + \omega_t + \epsilon_{i,t}$$

First stage (current)	Estimate
π_1	**1.567**
π_2 (Not S.S.)	0.068

Joint F: 10.1 (p=0.0065)
R^2 = 0.15

C. IV Results

Given the strong first-stage, I estimate the causal effects of new production using two-stage least squares at both the county and county-plus-100-mile level. The IV estimates produce positive and statistically significant coefficients, indicating that one million dollars of predicted new production is associated with an increase of $80,000 in BLS wages and $132,000 in IRS other income at the county level. These effects become substantially larger as I expand the geographic area: at the county-plus-100-mile level, the associated increases rise to $257,000 in BLS wages and $286,000 in IRS other income.

ΔY_{it} = Δ BLS Wages	Estimate
β_1	79,751 (9,394)
β_2 (Not S.S.)	-12,534 (7,505)

ΔY_{it} = Δ IRS other inc.	Estimate
β_1	132,016 (23,358)
β_2	-113,855 (32,988)

ΔY_{it} = Δ BLS Wages	Estimate
β_1	257,252 (20,455)
β_2 (Not S.S.)	-26,821 (17,045)

ΔY_{it} = Δ IRS other inc.	Estimate
β_1	286,000 (30,632)
β_2	-306.864 (52,235)

More importantly, these IV estimates imply that fracking shocks raise income not only in the wellhead county, but also in the broad surrounding region. The larger 100-mile

coefficients suggest that most economic gains occur outside the wellhead county, consistent with the idea that fracking generates regional spillovers through commuting, supplier networks, and service-sector linkages. Ultimately, the IV results provide the core causal evidence that a substantial portion of fracking's economic value accrues to the wider regional labor market rather than remaining exclusively at the extraction site.

While IRS other income captures royalty and lease payments accruing to mineral-rights holders and landowners, the authors note these payments go beyond royalties. DrillingInfo reports average royalty rates of about 12-15% of production value, implying that IRS other income also reflects other income channels like rental, business, and associated non-wage income linked to drilling. By summing BLS wages (labor-market earnings paid in the county) and IRS other income (royalties, landowner payments, business income accruing to residents), we obtain the total local income benefit. Using this combined measure, the IV estimates imply that 20% of the value of new production remains within the wellhead county and 54% remains within the surrounding 100-mile region.

D. IV Results by Industry

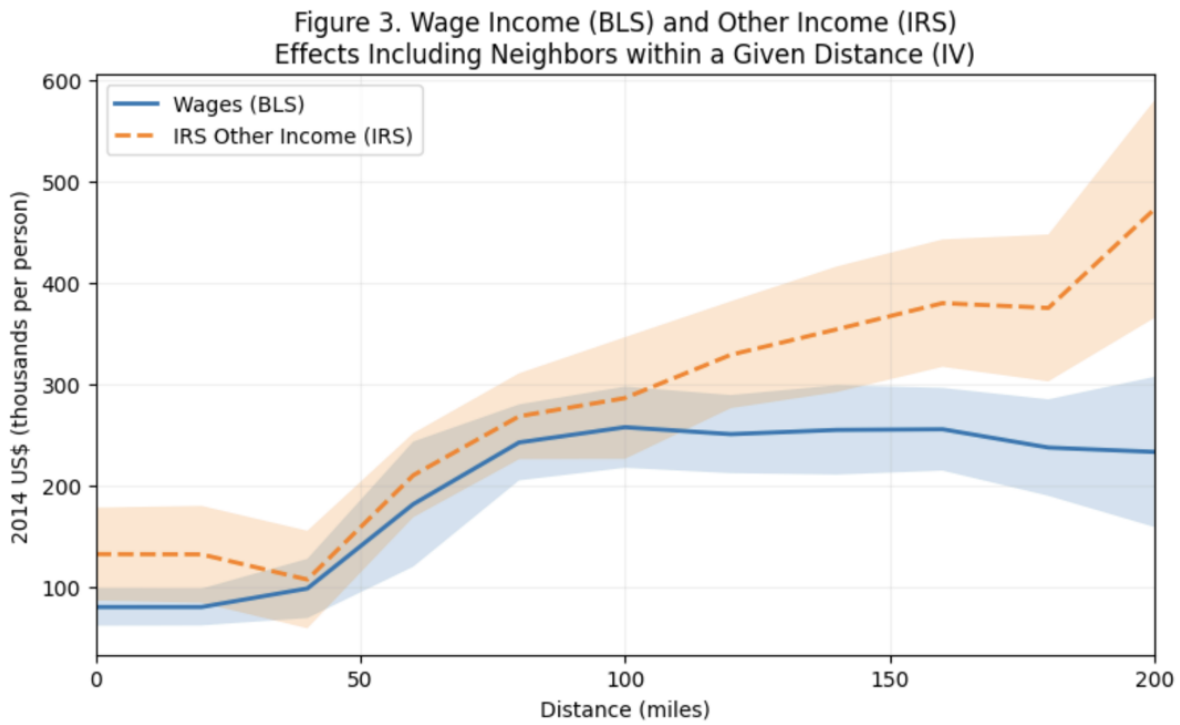
Finally, to examine how the income gains distribute across sectors, I estimate the industry-specific IV effects using the author's preferred 100-mile specification. The IV industry results indicate that a substantial portion of the immediate income gains accrues to extraction-related industries such as mining, transportation, and construction, with mining alone accounting for 60% of the contemporaneous income response. While the immediate effects are understandably concentrated in extraction-associated sectors, the dynamic event study presented later will show that income spillovers extend beyond extraction over time. Interestingly, there is a contemporaneous drop in the manufacturing industry, consistent with the resource-curse phenomenon, which is an ongoing debate in the current literature on energy shocks.

Total	Mining	Trans.	Const.	Manuf.	Ed./Health	Govt.	Oth. Serv.
257,252 (20,455)	155,886 (14,546)	61,374 (8,073)	36,201 (3,413)	-16,006 (3,256)	1,433 (1,651)	10,346 (6,822)	8,018 (3,471)

VI. Dispersion over Space and Time

A. Spatial Propagation

To assess how the economic benefits of new oil and gas production persist or dissipate over space, I replicate the original study's spatial propagation analysis. For each county, I construct regional income per worker and new production per worker within concentric radii around the county centroid and then re-estimate the IV model using the radius-based variables. The author's preferred 100-mile radius is not random: as the radius expands, additional counties enter the regional labor market, increasing both aggregate production and income response.

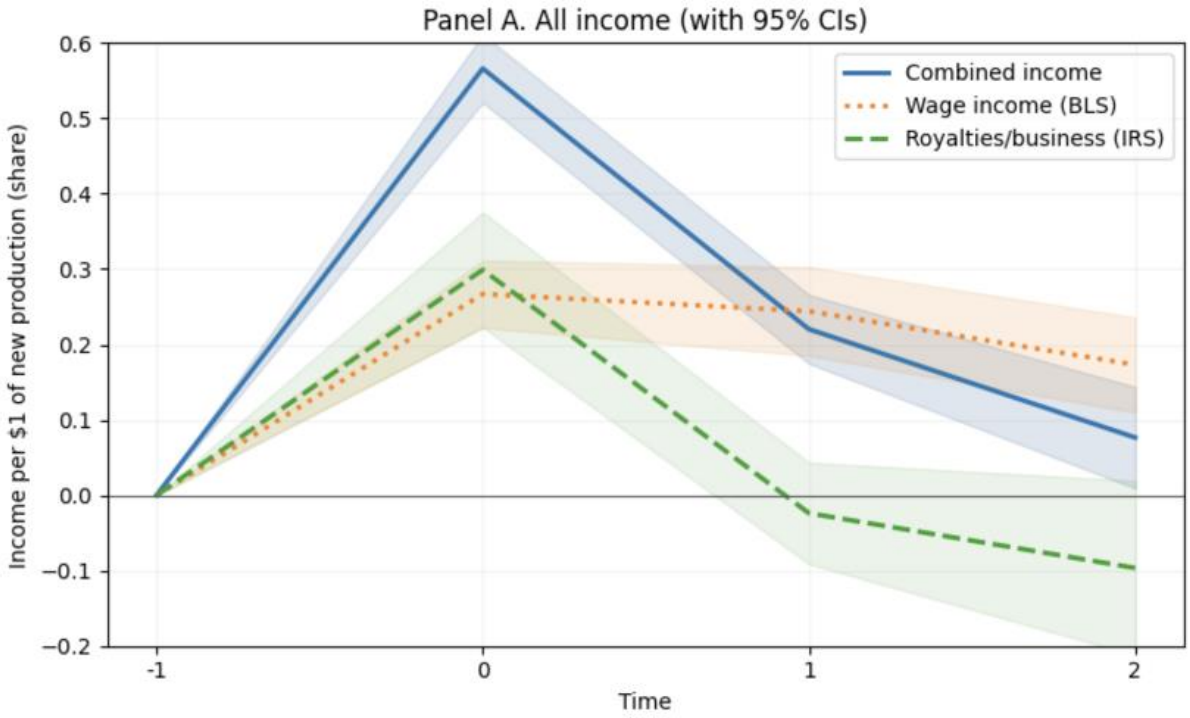


Between 0 and 50 miles, relatively few counties lie within the radius, so the IV coefficients closely match the county-level (0-mile) estimates. Between 50 to 100 miles, the coefficients increase considerably, with the effect on BLS wages reaching a $257,000 peak at the 100-mile margin. Beyond 100 miles, the BLS wage effect flattens, while IRS other income continues to rise substantially, suggesting that mineral-rights holders, landowners, and pass-through business owners typically live outside the wellhead county. These results

indicate that payroll gains are not confined to the drilling county but spill over into adjacent labor markets, and that most of the income effect is captured within 100 miles of the wellhead county.

B. Persistence

So far, the analysis has focused on contemporaneous impacts, but new oil and gas production may generate income effects that persist beyond the initial year of extraction. To examine how income effects evolve over time, I follow the authors and estimate the preferred 100-mile IV specification with two annual lags of new production. Although the limited time window restricts the event study, the coefficients still reveal some important patterns. Panel A traces the income response to a one-unit shock in new production, decomposed into BLS wage income, IRS other income, and their sum.

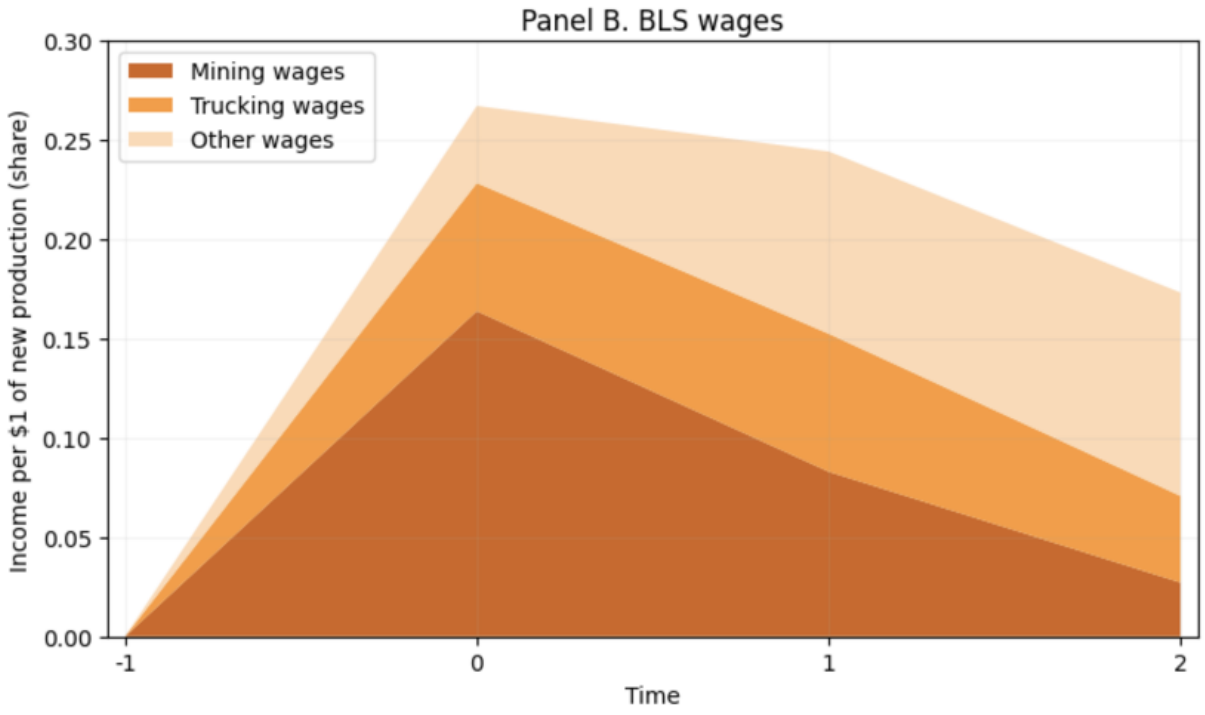


Although the BLS wage effect declines after the impact year, it remains significantly elevated for two full years following the shock, indicating that wage effects persist well beyond initial production. Consistent with the original paper, roughly two-thirds of the initial wage increase remains by year 2. This is because wells don't fully shutdown after the first year of production: they continue to produce at 40% of the initial peak after 1 year, and

25% after 2 years, reflecting continued labor demand related to operating wells, maintaining equipment, and transporting product.

IRS other income rises substantially, but virtually dissipates by the next year, implying that royalty payments and such are one-time ordeals. The aggregate income effect (combining BLS wages and IRS other income) declines significantly after year 1, but remains positive by year 2, suggesting that the fracking boom generates a temporary but not entirely fleeting boost to local income.

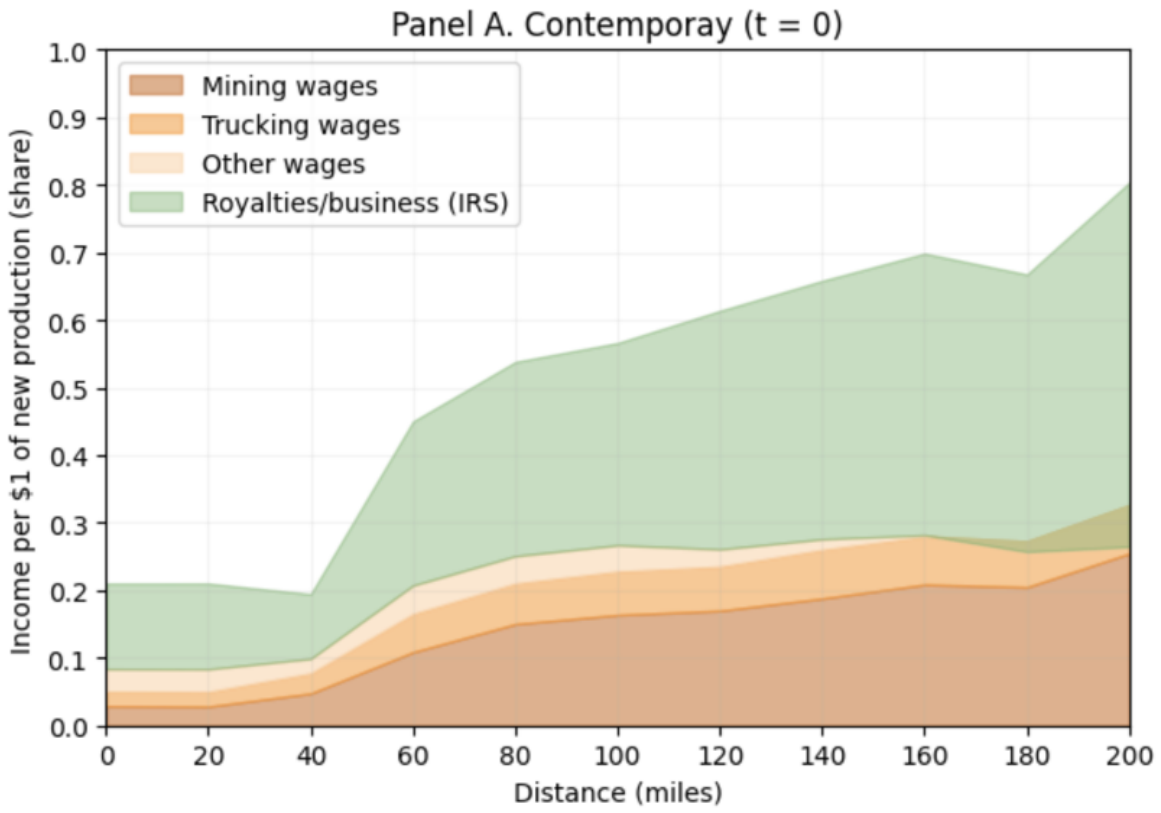
Panel B provides further insight by decomposing the BLS wage response into industries. Consistent with the IV industry estimates, mining and transportation account for most of the wage effects in the impact year, but mining gains fade quickly as drilling activity subsides. In contrast, wages in non-extraction sectors such as services, government, health, and education rise gradually over the following two-years, consistent with local multiplier effects and demand-driven service growth expansion as incomes rise.



All in all, the event study reveals a compositional shift in income effects: fracking initially generates large, but temporary gains in extraction-related earnings, followed by durable increases in non-related sectors. Thus, while the immediate boom is large, the subsequent effects rise more through spillovers and broader regional activity rather than continued drilling at the wellhead.

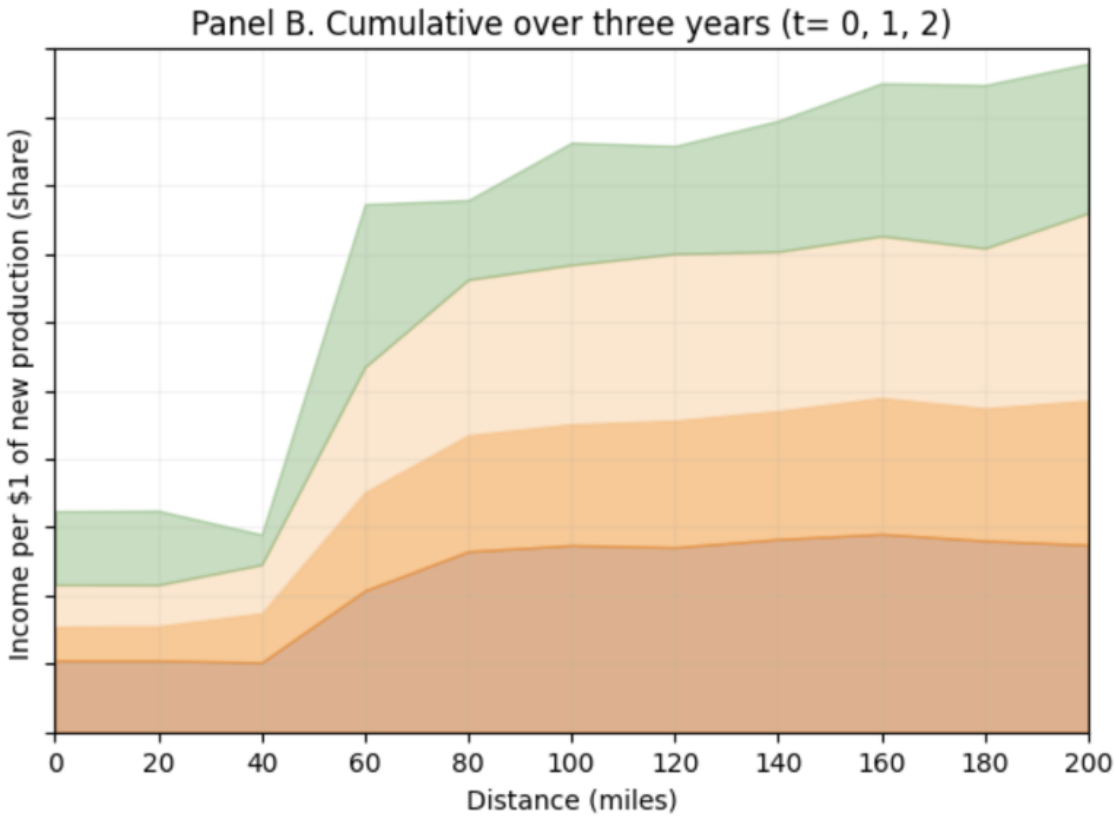
C. Effects over Space and Time

Finally, to study how the income effects evolve jointly across space and time, I first replicate Figure 5 Panel A, which plots the contemporaneous income effect across increasing radii, decomposed into industry wages and IRS other income. Consistent with our earlier results, the magnitude of the impact rises sharply as geographic radius widens, with IRS other income and mining wages accounting for the bulk of the effect of the immediate gains.



Panel B reports the cumulative income effect over the first three years following new production. The cumulative response becomes more evenly distributed across mining, transportation, other labor wages, and royalties/business income as activity spreads and spillovers occur. Within the drilling county (0-mile radius), the three-year cumulative effect equals one-third of the value of initial production, while the cumulative income effects 80% within a 100-mile radius. These findings highlight that the economic gains from fracking accrue primarily at the regional level rather than being confined to the wellhead county. As my professor noted under the presentation notes, “New York was correct to not allow

fracking and to gain the spillover effects from PA fracking within 100 miles of the NY border."



VII. Conclusion

This paper provides a comprehensive replication of Feyrer, Mansur, and Sacerdote's (2017) examination of the economic effects of the fracking/shale revolution on local and regional income. Using the authors' full dataset, I successfully reproduce their original results on income, spatial propagation, and dynamic effects. I do not replicate the employment analysis, instead focusing on income results, which alone offers a robust picture of how fracking generated economic value into local and regional economies.

Across all specifications, I find strong evidence that new oil and gas production generates substantial increases in local income. However, these gains are not confined to the wellhead county. While roughly 20% of the value of new production remains within the county in the year of drilling, approximately 54% accrues within a 100-mile radius. The immediate effects are split almost evenly between BLS wage income and IRS-measured

royalties and business income, reflecting both labor market and capital income channels. At the industry level, mining wages accounted for most of the contemporaneous gains, followed by other extraction-related sectors such as transportation and construction.

The event study further shows that IRS other income effects are short-lived, while roughly two-thirds of the BLS wage gains persist for two years, driven largely by non-mining industries such as transportation, services, government, health, and education. These patterns are consistent with both the steep production decline curve of fracked wells and with broader local multiplier effects as income circulates throughout the regional economy over time.

Finally, by combining the spatial and temporal income effects of fracking, I find that over a three-year horizon, counties retain roughly one-third of the value of new production, while the 100-mile region captures more than 80% of the initial production. This underscores the key takeaway that the economic footprint of energy shocks extends far beyond the drilling county itself, with regional labor markets, landowners, and service industries capturing the majority of long-run benefits.

REFERENCES

Allcott, H. and Keniston, D. (2014) 'Dutch Disease or Agglomeration? The Local Economic Effects of Natural Resource Booms in Modern America', **NBER Working Paper** No. 20508.

Black, D., McKinnish, T. and Sanders, S. (2005) 'The Economic Impact of the Coal Boom and Bust', **The Economic Journal**, 115(503), pp. 449–476.

Blanchard, O.J. and Katz, L.F. (1992) 'Regional Evolutions', **Brookings Papers on Economic Activity**, 1992(1), pp. 1–75.

Fetzer, T. (2014) 'Fracking Growth', **Working Paper**.

Feyrer, J., Mansur, E. T. & Sacerdote, B. (2017) Geographic Dispersion of Economic Shocks: Evidence from the Fracking Boom. **American Economic Review**, 107(4), 1313–1334.

Jacobsen, G.D. and Parker, D.P. (2016) 'The Economic Aftermath of Resource Booms: Evidence from Boomtowns in the American West', **The Economic Journal**, 126(593), pp. 1092–1128.

Maniloff, P. and Mastromonaco, R. (2014) 'The Local Economic Impacts of Unconventional Shale Development', **Working Paper**.

Michaels, G. (2011) 'The Long Term Consequences of Resource-Based Specialisation', **The Economic Journal**, 121(551), pp. 31–57.

Shoag, D. (2010) 'The Impact of Government Spending Shocks: Evidence on the Multiplier from State Pension Plan Returns', **Working Paper**.

Weber, J.G. (2012) 'The Effects of a Natural Gas Boom on Employment and Income in Colorado, Texas, and Wyoming', **Energy Economics**, 34(5), pp. 1580–1588.

Weber, J.G. (2013) 'The Effects of Energy Development on Local Labor Markets', **Working Paper**.